

Semantic-aware Implicit Neural Compression for Physics Simulations

Jessica Ezemba¹ James Afful² Mei-Yu Wang³

¹Carnegie Mellon University · ²Iowa State University · ³Pittsburgh Supercomputing Center



PROJECT PAGE

Introduction

- Physics simulations generate terabytes of data, blocking integration into ML pipelines.
- Lossless compression achieves low ratios; lossy codecs break conservation laws and physics structure.
- Heterogeneous meshes require domain-specific preprocessing, limiting reuse.

Key Challenge

Can we compress simulations *extremely* while preserving physical meaning, not just visual fidelity?

Physics Domains - The Well

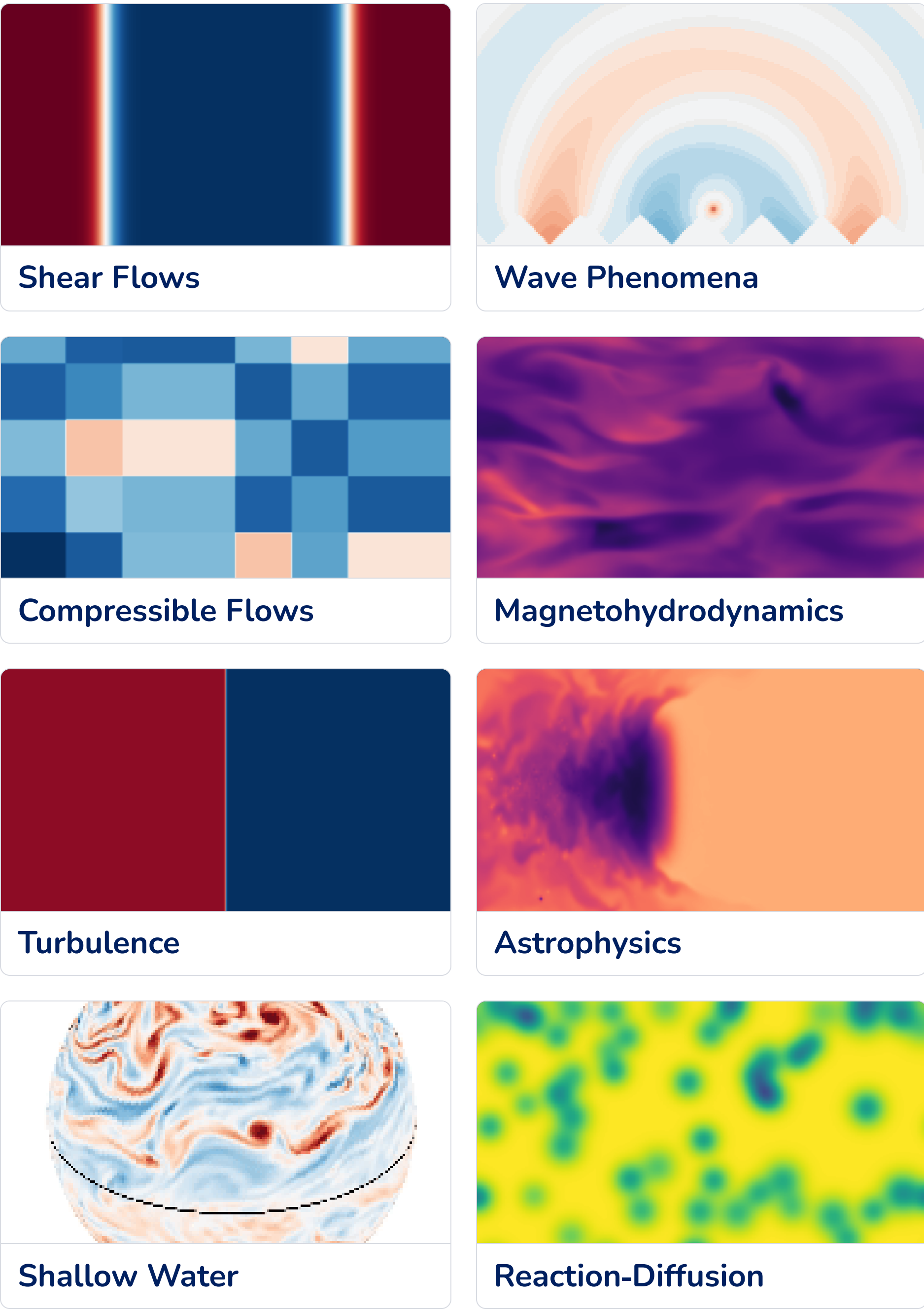


Figure 1. Representative sample of physics domains in The Well benchmark (22 datasets across 8 domains).

Objective

- Learn continuous **implicit neural representations (INRs)** of simulation data.
- Achieve extreme compression while preserving:
 - Conservation laws
 - Spectral structure
 - Multi-scale physical behavior
- Enable fast training and downstream ML use.

Selected Datasets

Domain	Class
Shear Flow	Rayleigh-Bénard (3), Shear Flow, Active Matter, Viscoelastic Instability
Wave Phenomena	Acoustic Scattering (4)
Compressible Flows	Euler (2)
Magnetohydrodynamics	Magnetohydrodynamics
Turbulence	Turbulence Gravity, Turbulent Radiative (2)
Astrophysics	Supernova Explosion (2), Hydrodynamic Radiation, Plasma Physics
Geophysical	Shallow Water
Reaction-Diffusion	Gray-Scott

Table 1. Description of the selected datasets.

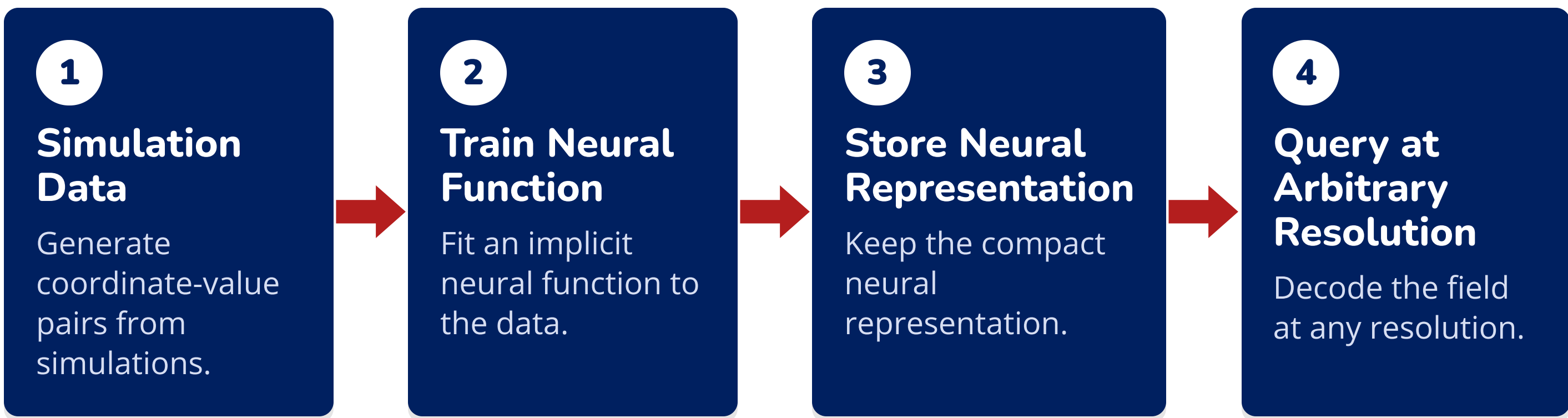
SINCPS Overview

Model

- SIREN** (Sinusoidal Representation Network) with Fourier positional encoding.
- Fourier encoding, 10 frequency levels (π to 512π).
- 4 hidden layers $\times$ 1024 units $\cdot$ $\sim$ 4.2M parameters.
- Input $(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{t}) \rightarrow$ field values $\cdot$ each saved model 37.6 MB.

Key Design Choices

- Static fields stored directly (float16).
- Dynamic fields encoded by the network.
- Per-field normalization for stability.



Traditional vs SINCPS

	Traditional	SINCPS
Representation	Discrete grid values	Continuous function
Resolution	Fixed at compression	Query at any resolution
Physics awareness	None	Conservation-law validation
Compression ratio	2 $\times$ to 4 $\times$ (lossless)	150 $\times$ to 25,000 $\times$

Results

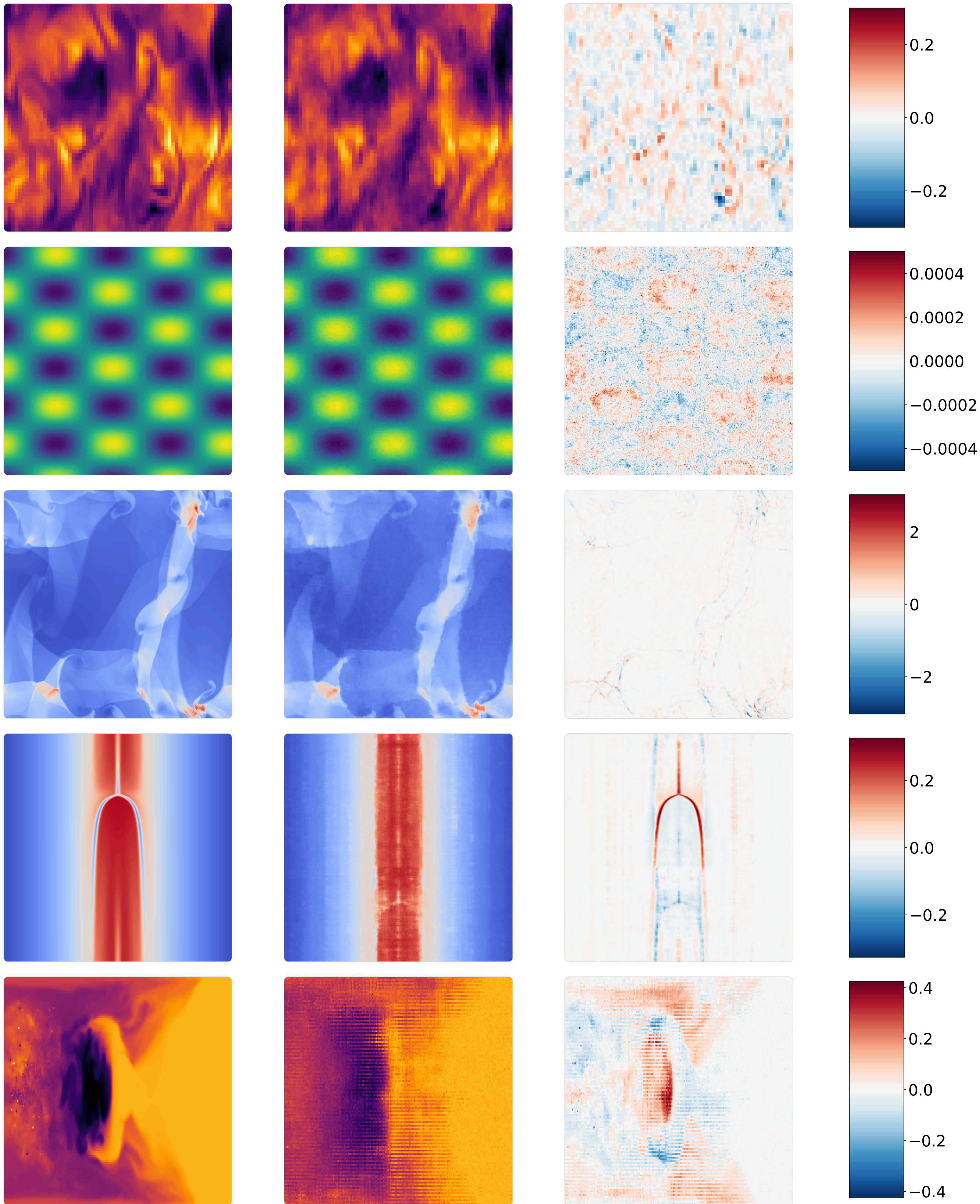


Figure 2. Reconstructed fields across physics domains, each row paired with its value colorbar.

Domain	Compression Ratio	PSNR (dB)	Spectral Error
Fluid Dynamics	1483 $\pm$ 780	21.41 $\pm$ 4.30	0.49 $\pm$ 0.40
Wave Phenomena	1750 $\pm$ 679	21.49 $\pm$ 8.74	1.32 $\pm$ 0.24
Compressible Flows	25348 $\pm$ 8	14.71 $\pm$ 5.97	1.21 $\pm$ 0.03
Magnetohydrodynamics	220	14.90	0.73
Turbulence	2394 $\pm$ 1988	13.69 $\pm$ 1.75	1.09 $\pm$ 1.17
Astrophysics	1767 $\pm$ 1338	32.25 $\pm$ 9.27	0.65 $\pm$ 0.31
Geophysical	182	14.08	0.62
Reaction-Diffusion	152	23.86	0.29

Table 2. Compression results showing standard and physics metrics. Initial size ranged from 5.4 GB to 1.9 TB; final compressed size is 37.6 MB.

METRICS DEFINED

- PSNR** $10 \cdot \log_{10}(\max^2/\text{MSE})$, in dB. >30 excellent $\cdot$ 20-30 good $\cdot$ <20 degraded. MSE = Mean Squared Error.
- Spectral error** Relative FFT error, $|\text{IFFT}(\text{pred}) - \text{FFT}(\text{true})| / |\text{IFFT}(\text{true})|$. Acceptable $< 15\%$.

Validation

- Conservation laws (mass, energy, $\nabla \cdot \mathbf{B}$).
- Spectral accuracy.
- Domain-specific diagnostics.

Training & Systems

- 22 datasets from The Well $\cdot$ one model per dataset.
- Per-dataset training time on the order of hours (hardware below).

Hardware Comparison - per dataset

Hardware	Time
Cerebras CS-3	2.3 h
GPU - NVIDIA H100	6.6 h
CPU - AMD EPYC 7702P	11.2 h

Physics Validation

Tiered Tolerances

- Conservation laws (primary): under 1%.
- Constitutive relations (secondary): under 5%.
- Statistical measures (L2, spectral, gradient): under 10 to 15%.

Per-domain Conservation Checks

Acoustic	wave speed, energy, impedance
Euler	mass, momentum, energy, shock speed
MHD	$\nabla \cdot \mathbf{B}$, mass, magnetic energy
Rayleigh-Bénard	incompressibility, Nusselt, heat flux
Shallow Water	mass, potential vorticity, energy
Supernova	shock radius, spherical symmetry, energy
Gray-Scott	reaction rate, diffusion balance

Conclusions & Future Work

What This Shows

- Implicit neural representations can serve as semantic compression.
- Physics constraints can be preserved at extreme compression.
- Per-dataset training runs in hours, not days.

Why It Matters

- Large simulation archives become queryable, shareable, and ML-ready.

Limitations

- 3D datasets need higher capacity for lower error.
- Training remains per-dataset.

Future Work

- Kolmogorov-Arnold Networks for high-frequency structure.
- Domain-specific physics validation.
- Neural operator training on compressed data.

References

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